

Project Report

PREDICTIVE ANALYTICS SYSTEM

GOOGLE MAPS BUSINESS ANALYTICS

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Table of Contents

1. Executive Summary
2. Introduction
3. Problem Statement
4. Objectives
5. Dataset Description
6. Data Collection
7. Data Cleaning & Preprocessing
8. Exploratory Data Analysis
9. Feature Engineering
10. Machine Learning Model
11. Model Evaluation
12. Dashboard Development
13. Business Insights
14. Opportunity Finder
15. Technologies Used
16. Future Improvements
17. Conclusion

1. EXECUTIVE SUMMARY

Overview:

The Google Maps Business Analytics Dashboard is an interactive web application developed using Python and Streamlit. The system allows users to upload any Google Maps business dataset, clean and preprocess the data, perform business analytics, generate visual insights, predict business opportunities using Machine Learning, and identify potential locations for business expansion.

The dashboard is designed for business owners, entrepreneurs, market researchers, and analysts who want to make data-driven decisions using publicly available Google Maps business information.

The project integrates data analytics, visualization, and predictive analytics into a single platform that supports interactive exploration and business intelligence.

2. INTRODUCTION

Business information available on Google Maps contains valuable insights regarding customer ratings, reviews, online presence, location, and business categories.

However, analyzing thousands of business records manually is time-consuming.

This project provides an automated solution that:

- Uploads Google Maps datasets
- Cleans the data
- Performs exploratory analysis
- Generates business insights
- Predicts business opportunities
- Visualizes results interactively

The application enables faster and more informed business decisions.

3. PROBLEM STATEMENT

Businesses often struggle to identify profitable locations, understand customer satisfaction, and evaluate competitors due to large volumes of unstructured data.

Manual analysis is inefficient and prone to errors.

There is a need for an intelligent system capable of:

- Processing Google Maps business data
- Cleaning inconsistencies
- Visualizing trends
- Predicting business opportunities
- Generating actionable insights

4. OBJECTIVES

The objectives of this project are:

- Build a system that analyses data.
- Accept any Google Maps business dataset.
- Perform automatic data preprocessing.
- Generate business insights.
- Visualize business distribution.
- Predict business opportunities using Machine Learning.

5. DATASET DESCRIPTION

The dataset contains business information collected from Google Maps.

Sample Attributes

Column	Description
Business Name	Name of business
Category	Business category
Rating	Customer rating
Reviews	Number of reviews
City	Business city
Area	Locality
Website	Website availability
Phone Number	Contact details
Address	Complete address
Latitude	Geographic latitude
Longitude	Geographic longitude

6. DATA COLLECTION

The dataset was collected from Google Maps business listings.

Data includes:

- Business details
- Customer ratings
- Review counts
- Geographic coordinates
- Contact information
- Website availability

The dashboard also supports uploading custom datasets in CSV format.

7. DATA CLEANING & PREPROCESSING

Several preprocessing techniques were applied:

Missing Values

- Filled missing ratings
- Removed invalid records
- Standardized missing values

Duplicate Removal

Duplicate businesses were removed.

Data Type Conversion

Converted:

- Ratings → Float
- Reviews → Integer
- Coordinates → Float

Text Cleaning

- Removed unwanted spaces
- Standardized city names
- Standardized categories

Feature Scaling

Numerical variables were prepared for machine learning.

8. EXPLORATORY DATA ANALYSIS

EDA was performed to understand the business landscape.

Key analyses include:

- Rating Distribution
- Average customer ratings.
- Category Analysis
- Most common business categories.
- Geographic Distribution
- Businesses across different cities.
- Website Analysis
- Businesses with and without websites.
- Review Analysis
- Businesses receiving the highest customer engagement.

9. FEATURE ENGINEERING

New features were created to improve prediction.

Examples include:

- High Rated Business
- Website Available
- Review Category
- Rating Category
- Opportunity Score
- Customer Engagement Score

These engineered features improve prediction accuracy.

10. MACHINE LEARNING MODEL

Algorithm Used

Random Forest Classifier

Reasons:

- Handles categorical variables effectively
- High prediction accuracy
- Resistant to overfitting
- Suitable for structured business datasets

Training Process

1. Data Encoding
 2. Train-Test Split
 3. Model Training
 4. Prediction
 5. Performance Evaluation
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11. MODEL EVALUATION

Evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

The trained model predicts whether a business has high growth potential based on available business attributes.

12. DASHBOARD DEVELOPMENT

The dashboard consists of the following modules:

1. Home
 - Overview and KPI cards.
 2. Business Overview
 - Business statistics.
 3. Rating Analytics
 - Rating trends and distributions.
 4. Geographic Analysis
 - Location-based visualizations.
 5. Online Presence
 - Website availability analysis.
 6. Customer Reviews
 - Review distribution.
 7. Prediction
 - Machine learning predictions.
 8. Opportunity Finder
 - Identifies business opportunities.
 9. Business Search
 - Search businesses by category or city.
 10. Data Explorer
 - Interactive dataset viewer.
 11. Model Explanation
 - Explains prediction methodology.
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13. BUSINESS INSIGHTS

The dashboard generates several insights such as:

- Cities with the highest business concentration.
- Categories receiving the highest ratings.
- Businesses lacking websites.
- Areas with low competition.
- High-growth opportunities.
- Customer engagement trends.

These insights support strategic business decisions.

14. OPPORTUNITY FINDER

The Opportunity Finder module identifies businesses with strong growth potential.

The prediction considers factors including:

- High customer ratings
- Large review counts
- Website availability
- Business category
- Geographic location

The model highlights businesses likely to perform well and areas where new businesses may succeed.

15. TECHNOLOGIES USED

Technology	Purpose
Python	Programming
Pandas	Data Analysis
NumPy	Numerical Computing
Streamlit	Dashboard Development
Matplotlib	Visualization
Scikit-learn	Machine Learning
Joblib	Model Saving

16. FUTURE IMPROVEMENTS

Potential enhancements include:

- Real-time Google Maps API integration
- Deep Learning models
- Business recommendation system
- Automated PDF report generation
- Sentiment analysis of reviews
- Competitor comparison
- Interactive maps using Folium
- Cloud deployment

CONCLUSION

The Google Maps Business Analytics Dashboard successfully combines data preprocessing, visualization, and predictive analytics into a single interactive platform.

The system enables users to upload business datasets, explore meaningful insights, identify growth opportunities, and make informed business decisions using machine learning.

This project demonstrates the practical application of data analytics, visualization, and predictive modelling in solving real-world business problems.